

Pathology Examination

Healing and Repair

Part I: Multiple Choice Questions (MCQs)

1. Regeneration is defined as replacement of damaged cells by:

- A) Fibrous scar tissue
 - B) New cells of a different kind
 - ✓ C) New cells of the same kind
 - D) Dense collagen fibers only
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2. Which cell type proliferates continuously throughout life to replace aging cells?

- A) Permanent cells
 - B) Stable cells
 - ✓ C) Labile cells
 - D) Muscle cells
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3. Stable cells are characterized by:

- A) Inability to divide under any circumstances
 - ✓ B) Dividing only when needed with variable regenerative capacity
 - C) Continuous rapid regeneration
 - D) Including nerve and cardiac muscle cells
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4. In fibrous repair, formation of vascular granulation tissue is called:

- A) Fibrogenesis
- ✓ B) Angiogenesis

- C) Maturation
 - D) Remodeling
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5. The first cells to infiltrate the wound and digest the clot are:

- A) Macrophages
 - B) Fibroblasts
 - ✓ C) Neutrophils**
 - D) Myofibroblasts
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6. Healing by secondary intention is characterized by:

- A) Surgically clean margins
 - B) Short healing time
 - C) Neat linear scar
 - ✓ D) Granulation tissue filling a large tissue gap**
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7. Which of the following is a general factor delaying wound healing?

- A) Foreign material such as sutures
 - ✓ B) Dietary deficiency of Vitamin C**
 - C) Poor local blood supply
 - D) Local infection or suppuration
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8. A keloid is best defined as:

- A) Blind-ending tract into tissues
- B) Abnormal communication between two surfaces
- ✓ C) Protuberant scar due to excessive fibrosis (overdone repair)**
- D) Obstruction of a tube or channel

Part II: Comparison – Primary vs Secondary Healing

Feature	Primary Healing	Secondary Healing
Cleanness	Clean	Unclean
Margins	Surgically clean	Irregular
Healing period	Short	Long
Healing direction	Direct	From bottom to edge
Outcome	Neat linear scar	Contracted irregular wound

Part III: Essay & Short Answer Questions

Q1. Factors Affecting Wound Healing

Local Factors:

- Type, size, and site of wound
- Movement or lack of apposition
- Infection (suppuration, gangrene, tetanus)
- Local blood supply
- Foreign materials (dirt, necrotic tissue)

General Factors:

- Age
- General health (e.g. diabetes)
- Drugs (e.g. steroids)

- Cardiovascular status
 - Dietary deficiencies (protein, Vitamin C, sulfur-containing amino acids)
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Q2. Complications of Repair

Insufficient Fibrosis:

- Ulceration
- Sinus (blind-ending tract)
- Fistula (abnormal communication between two epithelium-lined surfaces)

Excessive Contraction:

- Contractures (especially after burns)
 - Strictures (obstruction of tubes or channels)
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Q3. Epidermal (Implantation) Cyst

Definition:

A **keratin-filled cyst** caused by proliferation of **epithelial cells trapped deep in a wound** during regeneration.